SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I, R DINESH KUMAR (192312258), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: DINESH KUMAR R

ANNEXURE A — ANSWERS

Machine Learning | ITA 06 | Selected Questions

Question 1

A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is −0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Given: Study hours $x_1 = 2$; attendance $x_2 = 1$; weights $w_1 = 0.6$ and $w_2 = 0.4$; bias $b = -0.5$.

Net input: $z = w_1x_1 + w_2x_2 + b$

Substitution: $z = (0.6 \times 2) + (0.4 \times 1) - 0.5 = 1.2 + 0.4 - 0.5 = 1.1$.

Step activation: The step function returns 1 when $z \geq 0$ and 0 when $z < 0$. Since $z = 1.1 \geq 0$, the output is $y = 1$.

Conclusion: The perceptron predicts that the student passes.

Question 2

In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given: Initial weight $w = 0.5$; learning rate $\eta = 0.1$; input $x = 2$; actual output $y = 0.6$; target output $t = 1$.

Output error: $e = t - y = 1 - 0.6 = 0.4$.

Delta-rule weight update: Using $\Delta w = \eta \times e \times x$, $\Delta w = 0.1 \times 0.4 \times 2 = 0.08$.

Updated weight: $w(new) = w(old) + \Delta w = 0.5 + 0.08 = 0.58$.

Conclusion: After one training iteration, the updated weight is 0.58. The positive update moves the output in the direction of the target under the stated delta-rule calculation.

Question 3

A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Given: TP = 180; TN = 150; FP = 20; FN = 50. Total cases = 180 + 150 + 20 + 50 = 400.

Accuracy: Accuracy = (TP + TN) / Total = (180 + 150) / 400 = 330 / 400 = 0.825 = 82.50%.

Precision: Precision = TP / (TP + FP) = 180 / (180 + 20) = 180 / 200 = 90.00%.

Sensitivity (recall): Recall = TP / (TP + FN) = 180 / (180 + 50) = 180 / 230 ≈ 78.26%.

Specificity: Specificity = TN / (TN + FP) = 150 / (150 + 20) = 150 / 170 ≈ 88.24%.

Evaluation: The model has 90.00% precision, but its sensitivity is about 78.26%, meaning it misses approximately 21.74% of actual positive cases. In medical diagnosis, false negatives can delay treatment and cause harm. Therefore, these results alone are not enough to declare the model clinically reliable. It should be validated on representative clinical data, and sensitivity should be improved where missing a disease is especially dangerous.

Question 4

A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Fitness calculation: Fitness is $f(x) = x^2$. For x = 2, fitness = 4; x = 4, fitness = 16; x = 6, fitness = 36; x = 8, fitness = 64.

Selection: Because the objective is to maximize fitness, the best two candidates are x = 8 (fitness 64) and x = 6 (fitness 36).

Binary representation: Using four-bit representations: 8 = 1000 and 6 = 0110.

Crossover: Exchange the least significant (rightmost) bit: 1000 and 0110 both end in 0, so exchanging those bits produces offspring 1000 (8) and 0110 (6). In this particular population, the specified crossover does not change the selected individuals because both least significant bits are 0.

Role of mutation: Mutation randomly changes one or more bits or genes, introducing new genetic material into the population. It helps maintain diversity, reduces premature convergence, and may allow the search to explore better solutions. A suitable low mutation rate is generally used to avoid disrupting good candidates too often.

Question 5

During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

Observed trend: Training accuracy rises from 60% to 95%, while validation accuracy rises to 72% and then falls to 68%. The growing gap between the two indicates overfitting.

Why it occurs: After a certain point, the network learns details and noise specific to the training data rather than patterns that generalize to unseen data. Training performance continues to improve, but performance on validation data deteriorates.

Technique 1 — Early stopping: Monitor validation performance during training and stop when validation accuracy no longer improves or validation loss begins to rise. This avoids unnecessary training beyond the best generalization point.

Technique 2 — Dropout: Randomly deactivate a proportion of neurons during training. This discourages excessive dependence on particular neurons and can improve generalization.

Significance of validation accuracy: Validation accuracy estimates how well the model performs on data not used to update its weights. It helps compare models, tune hyperparameters, detect overfitting, and select a suitable stopping point. Final performance should also be checked on a separate test set.